The flow of talk or Agenda

FundaMental Of the SysteM

about the kernel

what Make the System Slow

is eBPF is necessary ? why it is

Difference Between eBPF and bpftrace

Why still eBPF is not yet Mature ?

There are no wrong answers. Speak up — you're already winning

List any software which you think it is Damm! slow i hate it

eg, any mobile application or
Operating system etc

Zara still uses MS DOS for their POS system

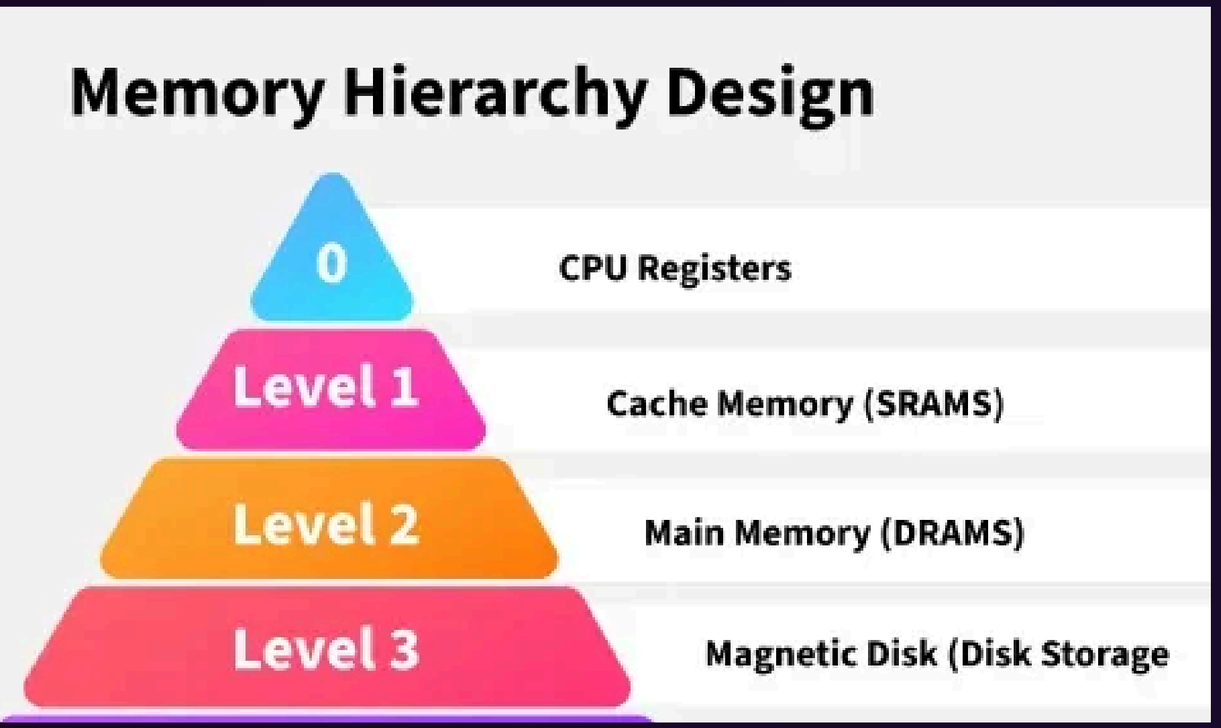
Zara IT for Fast Fashion Case Study

This case study aims to identify what operating system would be best for Zara. Should they maintain their current…

academia.edu

btw, Windows telemetry is actually a pretty advanced observability system. they mainly use ETW (Event Tracing for Windows) and even support DTrace-style tracing for deep system monitoring

FundaMental Of the SysteM



Main memory = unorganised

Secondary memory = organised

Is CPU can Predict the Future ?

CPU is Kinda astrologer

sometimes the prediction is correct, and other times it fails

modern CPUs do something called branch prediction to guess what the code will do next and keep execution fast. when the codebase grows or behavior changes, those predictions can fail and hurt performance, which is why tracing and profiling are super useful for finding bottlenecks

Source : https://blog.cloudflare.com/branch-predictor/

Source : https://en.wikipedia.org/wiki/Branch_predictor

daisyowl
@daisyowl

if you ever code something that "feels like a hack but it works," just remember that a CPU is literally a rock that we tricked into thinking

5:03 p.m. · 14 Mar 17

3,196 RETWEETS **5,051** LIKES

daisyowl @daisyowl · 17h

@daisyowl not to oversimplify: first you have to flatten the rock and put lightning inside it

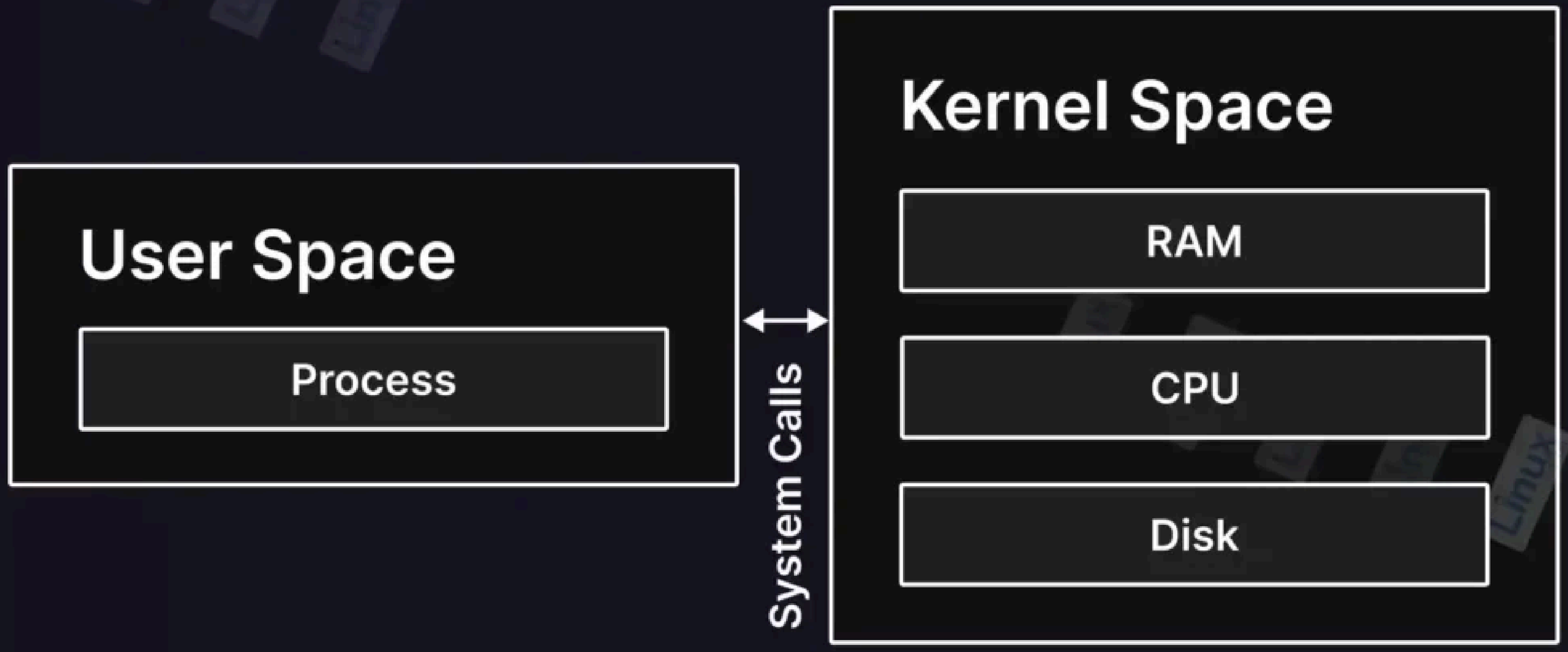
Posted in r/**programmingmemes** reddit

about the kernel

Kernel = Middle man between hardware and Software

Hardware Refers to CPU etc..

Software Refers to functionality of the system eg Browser ,


User Space
Process
System Calls
Kernel Space
RAM
CPU
Disk

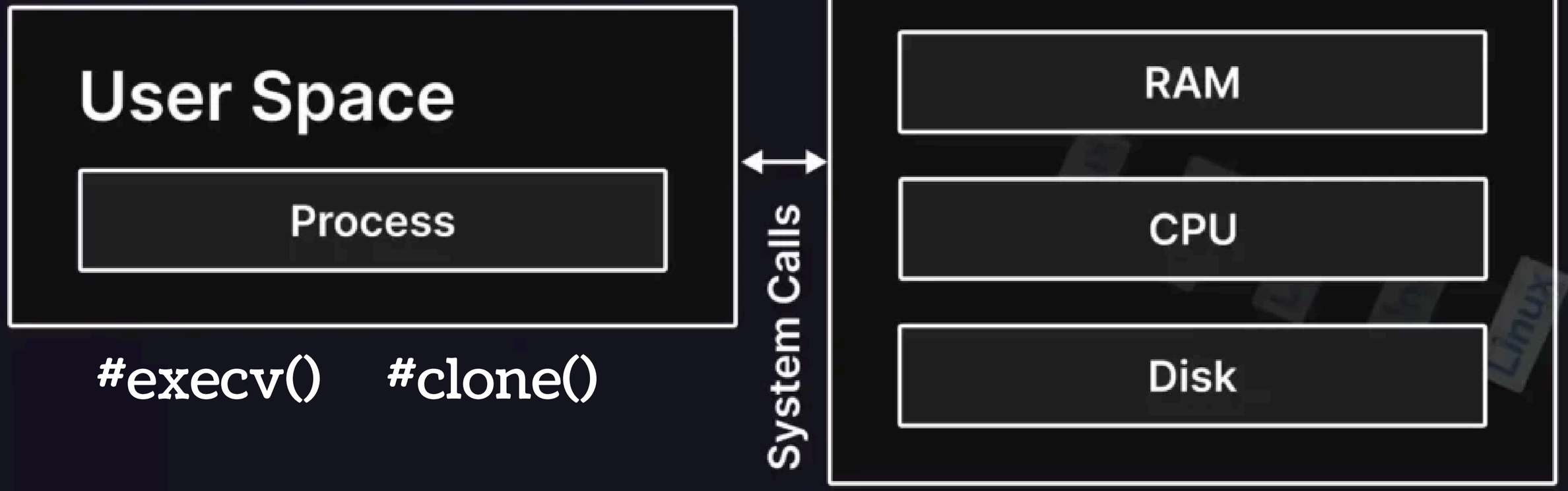
what Make the System Slow

Lemme Show a demo

is eBPF is necessary ? why it is

#shared kernel

#Dockerengine



#execv() #clone()

are eBPF and bpftrace are Different ?

eBPF is a engine

bpfTrace is the instrument cluster

But ebpf code also affect performance if the logic was wrong or overkilled

TracePoint

A tracepoint is an official place in the kernel designed for tracing

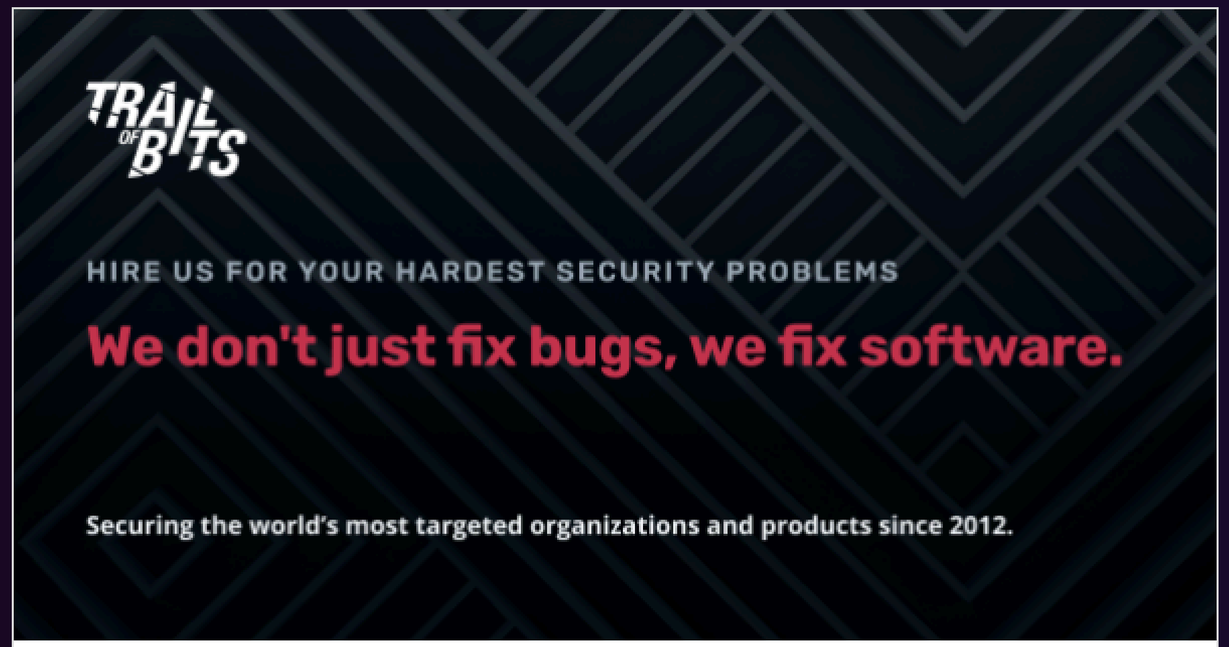
```
sudo bpftrace -e '
tracepoint:syscalls:sys_enter_execve
{
    printf("%s executed a program\n", comm);
}
'
```

Probes

a hook where tracing logic can attach and observe execution

```
sudo bpftrace -e '
kprobe:vfs_read
{
    printf("%s is reading a file\n", comm);
}
'
```

Why still eBPF is not yet Mature ?



Pitfalls of relying on eBPF for security monitoring (and some solutions)

eBPF (extended Berkeley Packet Filter) has emerged as the de facto Linux standard for security monitoring and endpoint observability. It is used by technologies such as BPFTrace, Cilium, Pixie, Sysdig, and Falco due to its low…

The Trail of Bits Blog / Sep 25, 2023

An Empirical Study on the Challenges of eBPF Application Development |…

As part of the Digital Library's transition to Open Access, new features for researchers…

ACM Conferences

Why performance matters

aws

annapurnalabs
an amazon company

Swiggy

So when the abstraction increases performance Decreases